Why do hotel bookings get cancelled?

Cancellation and revenue analysis of two Portuguese hotels, July 2015 to August 2017

119,390
bookings

2
hotels

8
questions answered

The answer: overbook in the City Hotel, not in the Resort

30% / 24%

of bookings are cancelled at the City Hotel / Resort Hotel. Cancellations are large enough to plan around.

97%

of non-refundable bookings are cancelled at the City Hotel, but they cancel early (median 100 days ahead), so the room can be resold.

+2 rooms

per night is a safe overbooking buffer for the City Hotel. The Resort Hotel should not overbook regularly.

Safe means: on at least 80% of past nights, enough bookings were cancelled in the last 7 days to cover the extra rooms.

From 119,390 raw bookings to 86,224 clean ones

1

Clean

1,301 contradictory records and 31,859 duplicates removed, plus 6 clear errors

2

Explore

Cancellation rates by month, lead time, deposit type, country and room change

3

Test

Chi-square and Mann-Whitney tests check which differences are real and not chance

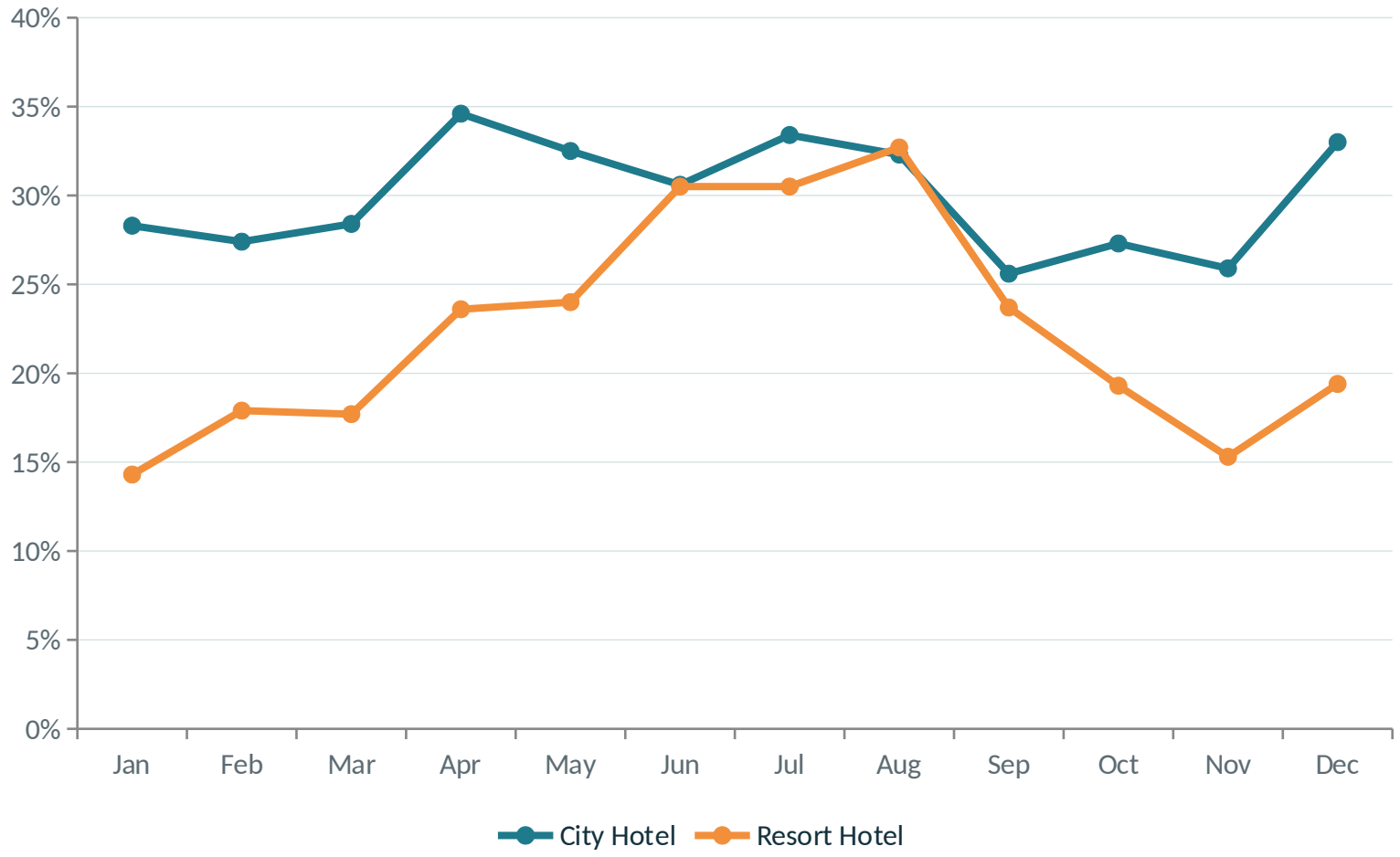
4

Recommend

Risk tiers and the timing of cancellations turn into an overbooking rule

Source: Antonio, de Almeida and Nunes (2019), via TidyTuesday, 2020-02-11

Cancellations peak in summer at the Resort Hotel



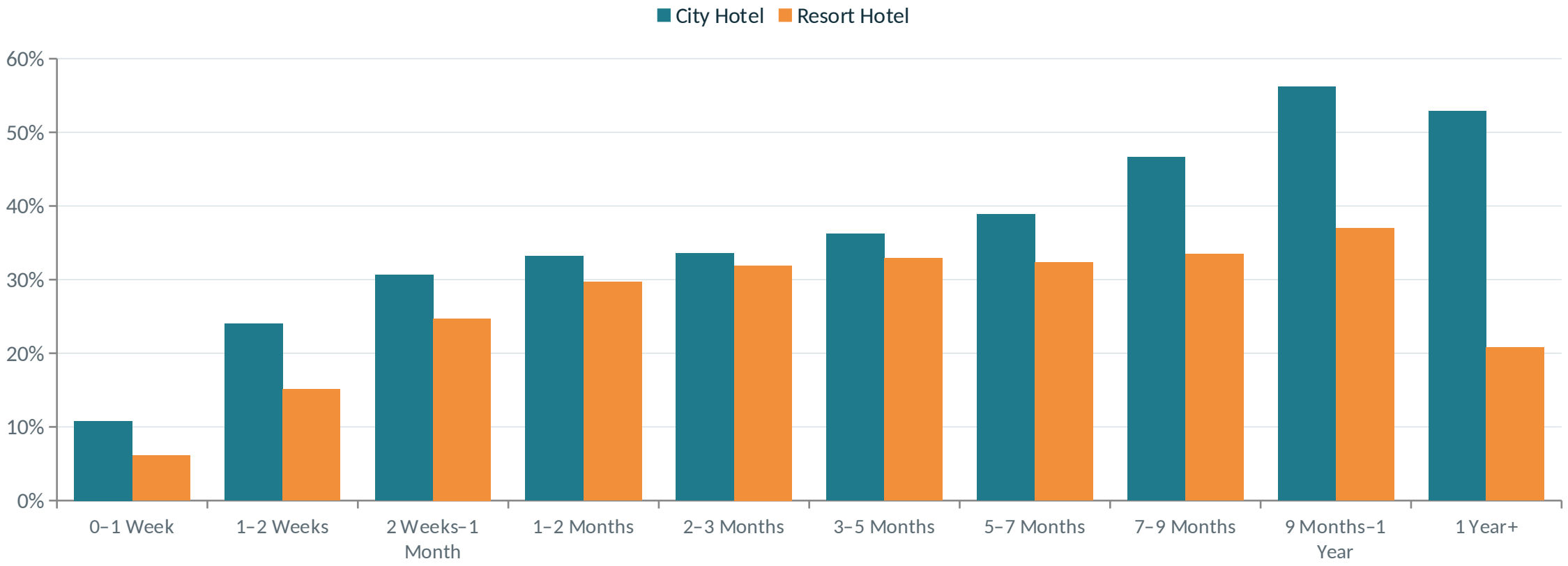
31.4% vs 19.8%

Resort Hotel cancellation rate in June to August vs the other months. The gap is real ($p < 0.001$).

34.6%

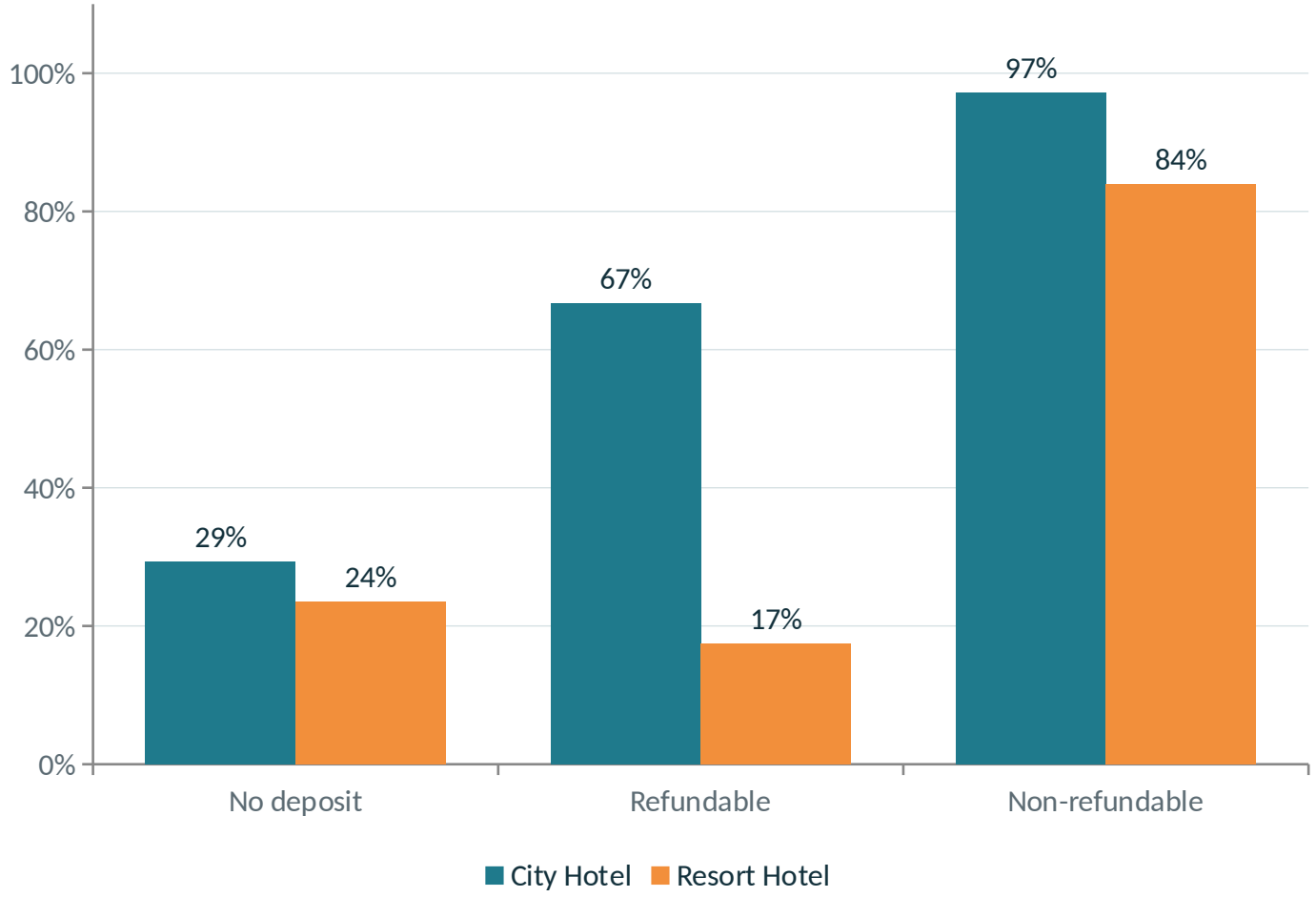
City Hotel peak, in Apr. Its rate stays above 25% in every month.

The earlier guests book, the more likely they cancel



In the City Hotel, 10.8% of bookings made in the last week are cancelled, against 56.2% of those made 9 months or more ahead.

Non-refundable bookings are cancelled almost every time



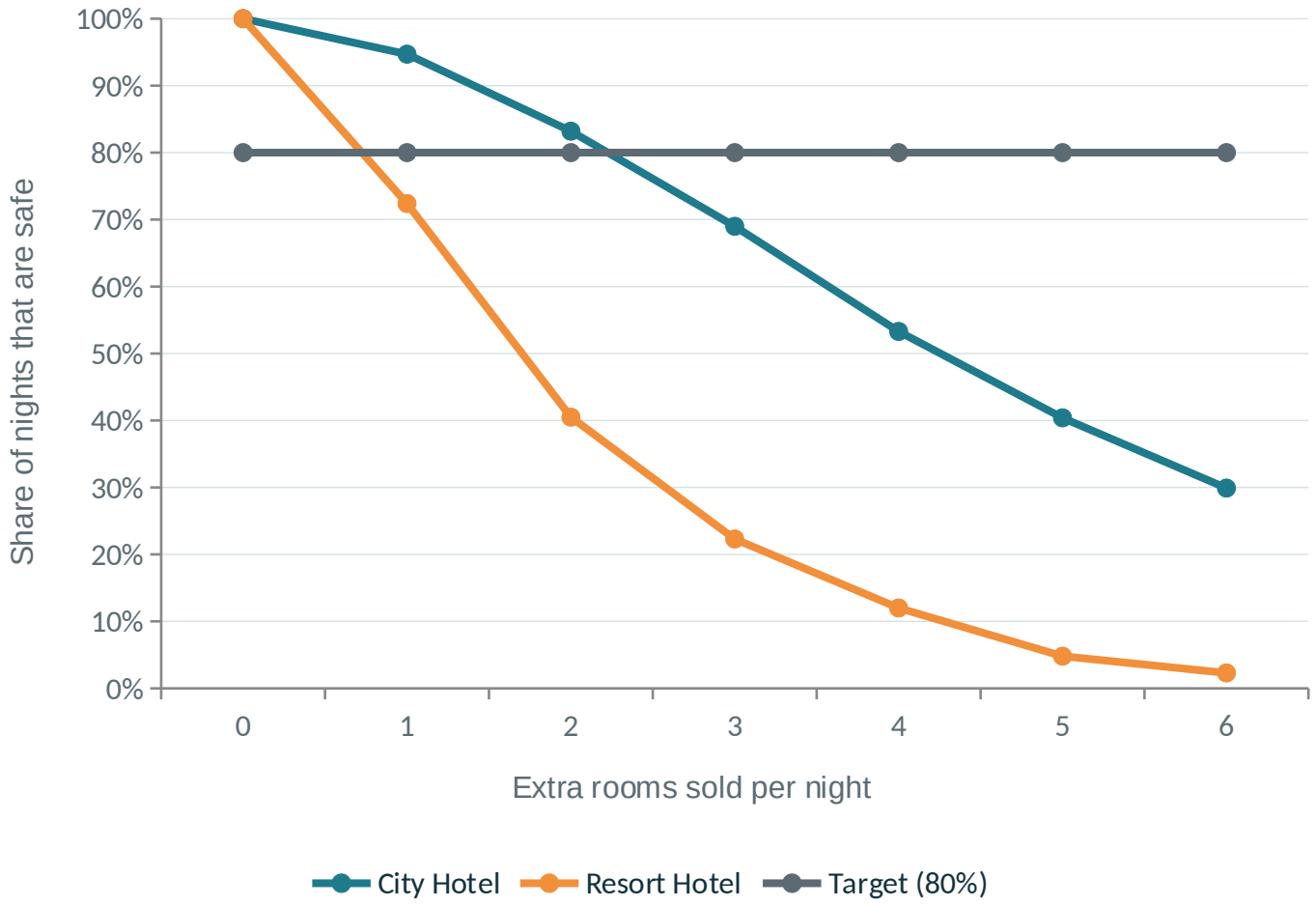
100 days

median time before arrival at which non-refundable bookings cancel (City Hotel)

5%

of their cancellations happen in the last week, so the room can be sold again

Recommendation: +2 rooms a night in the City Hotel



City: +2

rooms per night, about 3% of reservations. Safe on 83% of nights.

Resort: 0

No regular overbooking. One extra room is safe on only 72% of nights.

What the data cannot tell us, and what comes next

Limits

- Two hotels in one country, 2015 to 2017. Behaviour may have changed.
- No capacity or cost data, so the 80% target is a business choice.
- A late cancellation is defined as 7 days or fewer before arrival.
- Duplicate rows were removed. Without a booking ID, some may be real group bookings.

Next steps

- Build a model that predicts cancellations for each booking.
- Test the overbooking rule on newer data.
- Add the cost of walking a guest to pick the best target.

Appendix: how sure are we? Hypothesis tests

Question	Numbers	p-value	Conclusion
Resort: refundable deposits cancel less?	17.4% vs 23.9%	0.18	Not enough evidence
Resort: Jun to Aug cancel more?	31.4% vs 19.8%	< 0.001	Real difference
Resort: deposit types differ?	3 groups	< 0.001	Rates differ
City: top 2 countries differ in revenue?	Median 168 vs 324	< 0.001	Real difference

Chi-square tests for rates and Mann-Whitney for revenue. A p-value below 0.05 means chance is an unlikely explanation.